Rooted Triangle–Tree Extensions: Two Positive Families for the Chromatic-Polynomial Graphon Bound

Abstract

For a finite graph H, put

$$\Phi_H(p) = (1-p)^{v(H)}\chi_H\left(\frac{1}{1-p}\right),$$

with the value at $p = 1$ understood by continuity. We prove the bound $t(H, W) \geq \Phi_H(p)$, throughout the full required interval $p \geq 1/2$, for two infinite families. The first is obtained from a triangle by attaching any number of leaves to one triangle vertex. The second is obtained by attaching any number of pairwise disjoint two-edge tails to one triangle vertex. The proof is valid for every graphon and uses a weighted rooted-triangle inequality and a separate one-variable convexity argument. In the six-vertex catalogue the new cases are Atlas 34, 36, and 92.

1 Graphons and the two graph families

Let (Ω, μ) be a probability space. A graphon is a symmetric measurable function $W\colon \Omega^2 \to [0, 1]$. For a finite simple graph H, its homomorphism density is

$$t(H, W) = \int_{\Omega^{V(H)}} \prod_{uv\in E(H)} W(x_u, x_v) \prod_{u\in V(H)} \mathrm{d}\mu(x_u).$$

No injectivity is imposed on the maps represented by this integral. Write

$$p = t(K_2, W) \quad \text{and} \quad d(x) = \int_\Omega W(x, y)\,\mathrm{d}\mu(y).$$

Thus $0 \leq d \leq 1$ and $\int d = p$.

Fix a triangle on vertices o, a, b, with o distinguished as its root.

- For an integer $r \geq 0$, let L_r be obtained by adding vertices $u_1, \ldots, u_r$ and precisely the additional edges ou_i. Thus L_r is a triangle with r leaves having the same neighbour o.
- For an integer $s \geq 0$, let Q_s be obtained by adding vertices $y_1, z_1, \ldots, y_s, z_s$ and precisely the additional edges oy_i, y_iz_i. Thus Q_s is a triangle with s internally disjoint two-edge tails having the same initial vertex o.

In particular, $L_0 = Q_0 = K_3$, while L_1 is the paw graph.

Every new vertex belongs to a tree attached to the triangle. Once the triangle is properly coloured, each new vertex has exactly $x - 1$ choices after the colour of its parent has been fixed. Hence

$$\chi_{L_r}(x) = x(x-1)^{r+1}(x-2), \qquad v(L_r) = r + 3, \tag{1}$$

$$\chi_{Q_s}(x) = x(x-1)^{2s+1}(x-2), \qquad v(Q_s) = 2s + 3. \tag{2}$$

Both families have chromatic number 3. If $q = 1 - p$, direct substitution in the definition of Φ gives

$$\Phi_{L_r}(p) = q^{r+3}\frac{1}{q}\left(\frac{p}{q}\right)^{r+1}\left(\frac{2p-1}{q}\right) = p^{r+1}(2p-1), \tag{3}$$

$$\Phi_{Q_s}(p) = q^{2s+3}\frac{1}{q}\left(\frac{p}{q}\right)^{2s+1}\left(\frac{2p-1}{q}\right) = p^{2s+1}(2p-1). \tag{4}$$

These polynomial identities also hold at $p = 1$ by continuity.

2 A weighted rooted-triangle inequality

Define the rooted triangle density

$$\tau(x) := \int_{\Omega^2} W(x,y)W(x,z)W(y,z)\,\mathrm{d}\mu(y)\,\mathrm{d}\mu(z).$$

Lemma 2.1 (Weighted rooted Goodman inequality)**.** *Let W have edge density $p > 0$. For every integer $k \geq 0$,*

$$\int_\Omega d(x)^k\tau(x)\,\mathrm{d}\mu(x) \geq \frac{2p-1}{p}\int_\Omega d(x)^{k+2}\,\mathrm{d}\mu(x).$$

Proof. For $j \geq 0$, put

$$M_j = \int_\Omega d(x)^j\,\mathrm{d}\mu(x).$$

Let T_W be the integral operator with kernel W, and set

$$I_k = \int_\Omega d(x)^k\tau(x)\,\mathrm{d}\mu(x),$$
$$J_k = \int_{\Omega^2} d(x)^k W(x,y)d(y)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y) = \left\langle d^k, T_W d\right\rangle.$$

For $A, B \in [0,1]$ one has $AB \geq A + B - 1$. Apply this to $W(x,y)$ and $W(x,z)$, multiply by the nonnegative quantity $d(x)^k W(y,z)$, and integrate. Tonelli's theorem gives

$$I_k \geq 2J_k - pM_k. \tag{5}$$

Put $U = 1 - W$, let T_U be its integral operator, and write

$$u = T_U\mathbf{1} = 1 - d, \qquad q = 1 - p.$$

Since $T_U d = T_U(\mathbf{1} - u) = u - T_U u$, we have

$$T_W d = p\mathbf{1} - T_U d = d - q\mathbf{1} + T_U u.$$

Consequently,

$$J_k = M_{k+1} - qM_k + \left\langle d^k, T_U u\right\rangle. \tag{6}$$

We next prove

$$\left\langle d^k, T_U u\right\rangle \geq \int_\Omega d(x)^k u(x)^2\,\mathrm{d}\mu(x). \tag{7}$$

By the symmetry of U, twice the difference between the left- and right-hand sides of (7) equals

$$\int_{\Omega^2} U(x,y)\big(u(y)-u(x)\big)\big(d(x)^k-d(y)^k\big)\,\mathrm{d}\mu(x)\,\mathrm{d}\mu(y).$$

This integrand is nonnegative: $d = 1-u$, so u and $d^k = (1-u)^k$ are oppositely ordered. This proves (7). Since $u = 1-d$, equations (6) and (7) yield

$$\begin{aligned} J_k &\geq M_{k+1} - qM_k + \int d^k(1-d)^2 \\ &= M_{k+2} - M_{k+1} + pM_k. \end{aligned}$$

Using this in (5),

$$I_k \geq 2M_{k+2} - 2M_{k+1} + pM_k.$$

Finally,

$$\begin{aligned} &2M_{k+2} - 2M_{k+1} + pM_k - \frac{2p-1}{p}M_{k+2} \\ &\quad= \frac{1}{p}\left(M_{k+2} - 2pM_{k+1} + p^2M_k\right) \\ &\quad= \frac{1}{p}\int_\Omega d(x)^k(d(x)-p)^2\,\mathrm{d}\mu(x) \geq 0. \end{aligned}$$

The claimed inequality follows. □

3 A local inequality for two-edge tails

Set

$$A(x) := (T_W d)(x) = \int_\Omega W(x,y)d(y)\,\mathrm{d}\mu(y).$$

The quantity $A(x)$ is the rooted homomorphism density of a two-edge tail whose initial vertex is fixed at x. Notice that $0 \leq A \leq 1$ and

$$\int_\Omega A(x)\,\mathrm{d}\mu(x) = \int_\Omega d(x)^2\,\mathrm{d}\mu(x) \geq p^2. \tag{8}$$

The equality is symmetry and Fubini; the inequality is Jensen.

Lemma 3.1 (Pointwise rooted triangle bound)**.** *For almost every $x \in \Omega$,*

$$\tau(x) \geq \max\{2A(x) - p, 0\}.$$

Proof. The inequality $W(x,y)W(x,z) \geq W(x,y) + W(x,z) - 1$, multiplied by $W(y,z) \geq 0$ and integrated in y, z, gives

$$\tau(x) \geq 2\int_\Omega W(x,y)d(y)\,\mathrm{d}\mu(y) - p = 2A(x) - p.$$

Also $\tau(x) \geq 0$ by definition. Taking the larger of the two lower bounds proves the assertion. □

Lemma 3.2 (Tail convexity)**.** *Fix $p > 0$ and an integer $s \geq 0$. The function*

$$f_{p,s}(a) = a^s \max\{2a - p, 0\}, \qquad 0 \leq a \leq 1,$$

where $a^0 = 1$, is convex and nondecreasing.

Proof. For $s = 0$ this is the maximum of the two affine functions 0 and $2a - p$. Suppose $s \geq 1$. The function vanishes on $[0, p/2]$, while on $[p/2, 1]$ it is

$$2a^{s+1} - pa^s.$$

Its first derivative on the latter interval is

$$a^{s-1}\big(2(s+1)a - ps\big) > 0,$$

and its second derivative there is

$$sa^{s-2}\big(2(s+1)a - p(s-1)\big) \geq 0.$$

At $a = p/2$ the right derivative is $p(p/2)^{s-1} > 0$, whereas the left derivative is 0. Thus the one-sided derivatives increase across the joining point, proving both convexity and monotonicity on the whole interval. □

4 The positive theorem

Theorem 4.1 (Rooted triangle–tree extensions)**.** *Let W be an arbitrary graphon with edge density p. For every $p \in [1/2, 1]$ the following statements hold.*

1. *For every integer $r \geq 0$,*
$$t(L_r, W) \geq p^{r+1}(2p - 1) = \Phi_{L_r}(p).$$
2. *For every integer $s \geq 0$,*
$$t(Q_s, W) \geq p^{2s+1}(2p - 1) = \Phi_{Q_s}(p).$$

Since $\chi(L_r) = \chi(Q_s) = 3$, the interval $p \geq 1/2$ is exactly the full required interval

$$p \geq 1 - \frac{1}{\chi(H) - 1}.$$

Proof. Integrating out the r leaves of L_r, conditional on the image x of the root o, gives

$$t(L_r, W) = \int_\Omega d(x)^r \tau(x)\, \mathrm{d}\mu(x).$$

Lemma 2.1, followed by Jensen's inequality, gives for $p \geq 1/2$

$$\begin{aligned} t(L_r, W) &\geq \frac{2p-1}{p} \int_\Omega d(x)^{r+2}\, \mathrm{d}\mu(x) \\ &\geq \frac{2p-1}{p} p^{r+2} = p^{r+1}(2p-1). \end{aligned}$$

Here multiplication by $(2p-1)/p$ preserves the Jensen inequality because this factor is nonnegative on the stated interval.

For Q_s, integrating out its s disjoint two-edge tails gives

$$t(Q_s, W) = \int_\Omega \tau(x) A(x)^s \, \mathrm{d}\mu(x).$$

By Lemma 3.1, Lemma 3.2, Jensen's inequality, (8), and monotonicity of $f_{p,s}$,

$$\begin{aligned} t(Q_s, W) &\geq \int_\Omega f_{p,s}(A(x)) \, \mathrm{d}\mu(x) \\ &\geq f_{p,s}\left(\int_\Omega A(x) \, \mathrm{d}\mu(x) \right) \\ &\geq f_{p,s}(p^2). \end{aligned}$$

For $p \geq 1/2$ one has $p^2 \geq p/2$, and therefore

$$f_{p,s}(p^2) = p^{2s}(2p^2 - p) = p^{2s+1}(2p-1).$$

The target identifications are exactly (3) and (4). □

5 Endpoints and equality examples

At the threshold $p = 1/2$, both targets in (3)–(4) vanish, and the conclusion also follows directly from nonnegativity of homomorphism densities. Thus no division by a vanishing quantity occurs. At $p = 1$, the condition $0 \leq W \leq 1$ and $\int W = 1$ forces $W = 1$ almost everywhere, so both the homomorphism density and the target equal 1.

There are equality examples throughout the interval at all multipartite grid points. For an integer $k \geq 2$, let T_k be the balanced complete k-partite graphon. It has edge density $p_k = 1 - 1/k$, and for every finite graph H,

$$t(H, T_k) = \frac{\chi_H(k)}{k^{v(H)}} = \Phi_H(p_k).$$

In particular, this applies to every L_r and Q_s. For $k = 2$ it gives the threshold equality $0 = 0$.

6 Catalogue consequences

Corollary 6.1 (New six-vertex catalogue cases)**.** *The following previously open Atlas graphs satisfy $t(H, W) \geq \Phi_H(p)$ for every graphon W throughout the full required density interval $p \geq 1/2$:*

Atlas ID	*graph6*	*family member*	$\chi_H(x)$
34	`D@{`	L_2	$x(x-1)^3(x-2)$
36	`DJc`	Q_1	$x(x-1)^3(x-2)$
92	`E?Fw`	L_3	$x(x-1)^4(x-2)$

Their targets are, respectively,

$$p^3(2p-1), \qquad p^3(2p-1), \qquad p^4(2p-1).$$

Proof. Atlas 34 is a triangle with two leaves adjacent to the same triangle vertex, so it is L_2. Atlas 36 is a triangle with one two-edge tail, so it is Q_1. Atlas 92 is a triangle with three leaves adjacent to the same triangle vertex, so it is L_3. The graph6 codes give independent machine-readable identifiers. The conclusions follow from Theorem 4.1. □

7 Inputs and scope of the argument

The proof above is self-contained apart from Tonelli–Fubini and Jensen's inequality. In particular, it does not invoke the odd-cycle theorem, the pure-chordal theorem, clique–odd-cycle cases, universal-vertex lifting, an arbitrary join closure, or any counterexample calculation. The weighted rooted-triangle inequality also appears as an ingredient in the separate clique–odd-cycle note, but Lemma 2.1 proves it here in full.

All arguments use the ordinary (non-injective) graphon homomorphism integrals directly. They require neither a finite-step approximation nor regularity of W, and they never replace the average edge-density hypothesis by a minimum-degree or pointwise degree assumption.

A What the Lean formalization actually proves

Corollary 6.1 is formalized: Atlas 34, 36 and 92 are `verified`, as is L_1 itself, the paw, which is Atlas 15 and which several later notes consume. The development is `lean/Taeyoung/Methods/RootedTriangleTree/`. One difference of substance.

Lemma 3.2 is not formalized

The piecewise function $a \mapsto a^s \max\{2a - p, 0\}$ does not occur, and neither family goes through it.

For the leaf family L_s the density factors as $\int d^s \tau$, and Lemma 2.1 sends that to $\frac{2p-1}{p} \int d^{s+2}$ directly; all that is then needed is $M_j \geq p^j$, which is Jensen for the convex $t \mapsto t^j$ against the degree function. The maximum with 0 never enters, because the weighted inequality has already been applied.

For the tail family Q_s, whose density is $\int \tau \cdot A$ with $A = T_W d$ and which does not factor as $\int d^r \tau$, the piecewise Jensen argument is replaced by algebra. The pointwise bound $\tau \geq 2A - p$ of Lemma 3.1 and $A \geq 0$ give

$$\int \tau A \;\geq\; 2\int A^2 - p\int A \;=\; 2M_2^2 - pM_2 + \Big(\int A^2 - M_2^2\Big) \;\geq\; 2M_2^2 - pM_2,$$

the last step being Cauchy–Schwarz, and $m \mapsto 2m^2 - pm$ is increasing beyond $m = p^2$ when $p \geq 1/2$, so the bound descends to $2p^4 - p^3 = p^3(2p - 1)$. No convexity of a piecewise function, and no one-sided derivatives at a joining point.